

UNIT 2

Summary



ESSENTIAL QUESTION

What happens to the energy and atoms of substances in chemical reactions?

TOPIC 2.1:

How are chemical processes part of our lives?

- Applications of chemistry are everywhere in the world around you.
- Knowing how to handle chemicals helps keeps us and our environment safe.

Key Term

chemical reaction



TOPIC 2.2:

What happens to atoms in a chemical reaction?

- Atoms bond together to form ionic and covalent compounds.
- Bonds are broken, atoms are rearranged, and new bonds are formed.
- Mass cannot be created or destroyed in a chemical reaction.
- A chemical equation represents what happens to the atoms in a reaction.

Key Terms

ionic compound

molecule

reactant

ionic bond

law of conservation of mass

product

covalent compound

chemical equation

coefficient

covalent bond



TOPIC 2.3:

How is energy involved in chemical processes?

- Matter and energy interact in physical and chemical changes.
- Energy is transferred between chemical reactions (the system) and the surroundings.

Key Terms

exothermic reaction
endothermic reaction



TOPIC 2.4:

How do atoms rearrange in different types of chemical reactions?

- A compound forms in a synthesis reaction and breaks down in a decomposition reaction.
- In replacement reactions, elements replace other elements.
- Most combustion reactions release heat and light.
- In a neutralization reaction, an acid reacts with a base.

Key Terms

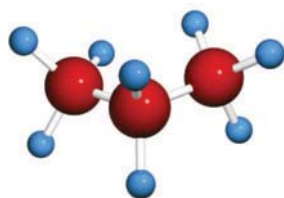
synthesis reaction	acid
decomposition reaction	base
single replacement reaction	acid-base indicator
double replacement reaction	pH scale
combustion reaction	neutralization reaction

Review

What Do You Know? Connecting to Concepts

Visualizing Ideas

1. A ball-and-stick model of propane, C_3H_8 , is shown. Use the model to answer the following questions.



- Is this an ionic or covalent compound? Explain how you know.
- Using Bohr or Lewis diagrams, show how electrons are involved in the formation of one of the bonds in this compound.
- Describe the bonds that are broken and new bonds that form when propane undergoes a combustion reaction.
- Sketch an energy level diagram that would represent the combustion of propane. Label your diagram as endothermic or exothermic, and provide evidence that supports your decision.

2. For each image below, identify the reaction type it represents, and explain why it represents that type of reaction.

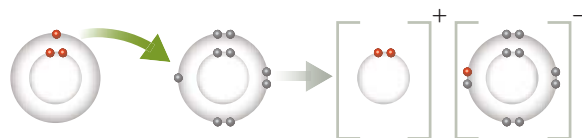


Using Key Terms

- What does the term *formula unit* refer to?
- What do coefficients in balanced chemical equations represent? Give an example of a chemical formula with a coefficient.
- Describe, in sentences or with a graphic organizer, the relationships between the following terms: *bond breaking*, *bond forming*, *energy release*, *energy absorption*, *exothermic*, *endothermic*.
- Describe what an acid and a base are. Give the name and chemical formula for an example of each.
- Use a graphic organizer to compare and contrast ionic and covalent bonds.

Communicating Concepts

- Determine the number of atoms of each element for the following compounds.
 - $4CO$
 - $3MgCl_2$
 - $2(NH_4)_2SO_4$
- Why is it incorrect to change the subscripts in chemical formulas when you are balancing a chemical equation?
- Balance each of the following chemical equations.
 - $Na(s) + O_2(g) \rightarrow Na_2O(s)$
 - $KCl(s) \rightarrow K(s) + Cl_2(g)$
 - $N_2(g) + O_2(g) \rightarrow NO_2(g)$
 - $CuSO_4(aq) + Na(s) \rightarrow Cu(s) + Na_2SO_4(aq)$
 - $Pb(NO_3)_2(aq) + KI(aq) \rightarrow PbI_2(s) + KNO_3(aq)$
 - $Al(s) + Cl_2(g) \rightarrow AlCl_3(s)$
- Use the diagram to answer the questions below.



- Two atoms are shown. What elements are represented? Explain how you know.
- What type of bond is represented? Why?
- Is energy released or absorbed in this process?

12. Why are ionic compounds not considered molecules?
13. Draw a Bohr diagram that represents the chemical bond in a molecule of Cl_2 .
14. What elements exist as diatomic or polyatomic molecules? Write their names and chemical formulas.
15. Indicate if each of the following is an acidic, basic, or neutral solution.
 - a) lemon juice
 - b) a solution with $\text{pH} = 9$
 - c) an aqueous solution of sodium chloride, using water with a $\text{pH} = 7$
16. Bacteria are used to make cheese ($\text{pH} = 5.5$) and yogurt ($\text{pH} 4.5$) from milk ($\text{pH} 6.5$). Place these foods in the order of least acidic to most acidic.
17. The photo shows what happens when zinc metal is placed in a solution of hydrochloric acid.
 - a) What evidence suggests that a chemical change is occurring?
 - b) How does the mass of zinc metal change as the reaction proceeds?
 - c) Name the gaseous product that forms bubbles in this reaction. How do you know?
 - d) Write the balanced chemical equation for this reaction.
 - e) What type of reaction is this? Explain your choice.
 - f) After the reaction, the test tube feels warm. Describe the overall energy change for this reaction.
18. You have learned about single replacement and double replacement reactions in this unit. Explain why the names of these types of reactions are appropriate.



What Do You Know?

Connecting to Competencies

Developing Skills

19. Represent these reactions using word, skeleton, and balanced chemical equations. Include the states of reactants and products.
 - a) Freshly cut sodium reacts with oxygen gas in air and forms solid sodium oxide.
 - b) When a piece of magnesium ribbon is placed into an aqueous solution of copper(II) chloride, copper metal and an aqueous solution of magnesium chloride form.
 - c) Solid magnesium oxide and carbon dioxide gas form when powdered magnesium carbonate is heated.
 - d) When aqueous solutions of chromium(III) chloride and potassium hydroxide are mixed, a solution of potassium chloride and a precipitate of chromium(III) hydroxide form.
20. Identify the type of reaction for each part in question 19. Explain your reasoning.
21. For the following reactants, identify the type of reaction they will undergo, predict the products, and write the balanced chemical equations. (Do not include the states of the products.)
 - a) $\text{Au}(\text{NO}_3)_3(\text{aq}) + \text{Ag}(\text{s}) \rightarrow$
 - b) $\text{CuO}(\text{s}) \rightarrow$
 - c) $\text{BaCl}_2(\text{aq}) + \text{K}_2\text{SO}_4(\text{aq}) \rightarrow$
 - d) $\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow$
 - e) $\text{HBr}(\text{aq}) + \text{Al}(\text{OH})_3(\text{aq}) \rightarrow$
 - f) $\text{AgCl}(\text{s}) \rightarrow$
 - g) $\text{Ca}(\text{s}) + \text{S}_8(\text{s}) \rightarrow$
 - h) $\text{Mg}(\text{s}) + \text{HCl}(\text{aq}) \rightarrow$
 - i) $\text{NaCl}(\text{aq}) + \text{AgNO}_3(\text{aq}) \rightarrow$
 - j) $\text{C}_4\text{H}_{10}(\ell) + \text{O}_2(\text{g}) \rightarrow$

Unit 2 Review *(continued)*

Thinking Critically and Creatively

- 22.** An element in Group 1 on the periodic table forms ionic compounds with elements in Group 17 in a 1:1 ratio. In what ratio would you expect an element from Group 2 and elements from Group 16 to react when they form ionic compounds? Explain your answer.
- 23.** In this unit, pictorial representations were used to help you understand what happens to elements and compounds in the different types of chemical reactions. These representations used coloured spheres. Develop a different method to represent what happens in the types of chemical reactions you have learned about. Be creative—for example, you might consider using cartoons of different people dancing.
- 24.** Draw a sketch that represents how energy can be exchanged between a system and its surroundings.
- For chemical reactions, what represents the system and what represents the surroundings?
 - If there is a net absorption of energy by a chemical reaction, what does that tell you about the energy of the surroundings?

What Do You Know? Making New Connections

Applying Your Understanding

- 25.** Develop an image that represents energy changes that occur during a chemical reaction. Your image should include the following labels: endothermic, exothermic, bond formation, bond breaking, reactant, product, and energy.
- 26.** The photo below shows wood burning.
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- What type of chemical reaction is this?
 - Is it exothermic or endothermic? What evidence supports your answer?
 - Does this reaction require an input of energy to get started? How do you know?
- 27.** When wood burns completely, a pile of ash remains. The mass of the ash is much less than the mass of the original wood.
- Does this observation invalidate the law of conservation of mass? Why or why not?
 - Describe an experiment that would provide evidence to support your answer to part a).
- 28.** Think of five ways you rely on chemistry in your daily life.
- Discuss the advantages and the disadvantages that are associated with using the chemicals.
 - What alternatives are available that might counter the disadvantages?
- 29.** In a highly exothermic reaction, solid carbon dioxide (dry ice) sublimates and reacts with hot magnesium to produce solid magnesium oxide and solid carbon.
- Fire extinguishers commonly contain carbon dioxide, which is heavier than air and smothers a fire. How does the chemical reaction described above demonstrate the limited usefulness of carbon dioxide fire extinguishers?

Thinking Critically and Creatively

- 30.** A chemist performs an experiment that involves reacting 1.6 g of sodium carbonate with 1.1 g of calcium chloride in an aqueous solution.
- What type of chemical reaction is this? Describe what is happening to the ions during the reaction.
 - Predict the products of this reaction.
 - Write the word equation, skeleton equation, and balanced chemical equation for this reaction. One product is not soluble in water and forms a precipitate. The other product is soluble in water.
 - The precipitate that forms is isolated using a filtration apparatus, shown here. The filter paper and solid are dried overnight. The next day the mass of the solid product is determined to be 1.0 g. What is the mass of the second product expected to be? How do you know?
 - The chemist expected to isolate a greater amount of precipitate. Describe one thing that might have contributed to the lower amount actually obtained.
- 31.** The photo shows a pH meter is being used to measure the pH of a solution.

- The number displayed represents the pH of the solution. Is it acidic or basic?
- Should HCl(aq) or KOH(aq) be added to neutralize this solution? Explain your reasoning.



Connecting to Self and Society

- 32.** Baking soda, sodium hydrogen carbonate, can be used to neutralize an acid spill. The chemical equation is
- $$\text{HCl(aq)} + \text{NaHCO}_3\text{(s)} \rightarrow \text{NaCl(aq)} + \text{CO}_2\text{(g)} + \text{H}_2\text{O(l)}$$
- Will the pH increase or decrease by adding baking soda? Explain.
 - Why is baking soda a safer alternative to other bases, such as NaOH(aq) ?
 - Write the balanced chemical equation for the neutralization of hydrochloric acid with magnesium hydroxide.
- 33.** One way to treat a body of water polluted with acid precipitation is to add calcium hydroxide, Ca(OH)_2 .
- How does this help?
 - In what way is liming a lake similar to taking an antacid for heartburn?
 - Why is lake liming a short-term solution to the problem?
 - Describe a longer-term solution to the problem of acidic lakes due to acid precipitation?
- 34.** While working in a laboratory as a summer student, you notice that the Safety Data Sheets for many of the chemicals cannot be found. What could you say to your supervisor to help convey the importance of maintaining up-to-date SDS records?
- 35.** What decisions about chemical products and processes do you make, personally, on a daily basis? What are the benefits and risks associated with your decisions? How can an understanding of chemistry help you better assess the benefits and risks?

Unit Assessment

How do chemicals affect agriculture in B.C.?



B.C.'s diverse geography and climates support a wide variety of fruits, vegetables, and grains, as well as livestock. The physical and chemical conditions of the soil of a given region help us decide what can be grown and raised. Adding chemicals such as fertilizers, elemental sulfur, and lime can enhance soil structure and nutrient content, as well as maintain its fertility. Understanding the properties of these chemicals and how they interact with their surroundings helps to promote sustainable yields and can reduce risks to society and the environment.

Work as part of a group to do the following.

- STEP 1** ▶ Reflect on the three options, their photos, and the questions asked in their captions.
- STEP 2** ▶ Brainstorm at least three more options and questions of your own about chemicals in agriculture.
- STEP 3** ▶ Decide on one of the six questions to investigate.
- STEP 4** ▶ Plan and conduct a scientific inquiry to explore your question.
- STEP 5** ▶ Organize and analyze the data and information that you find and collect.
- STEP 6** ▶ Communicate the results of your inquiry in a suitable manner.

OPTION A

Agricultural Chemistry

What chemicals are used for growing crops in the field and in hydroponics to modify pH conditions, and for what reasons?



OPTION B

Crops around the World

How do agricultural chemicals affect the stability, fertility, and productivity of the soil in different geographical locations?



OPTION C

Impacts of Human Activity

What chemical reactions naturally occur in soil ecosystems, how does human agricultural activity alter them, and with what consequences?



Assessment Criteria

Did I and my group...

- Develop one or more questions that provided opportunities for rich investigation? **QP**
- Develop effective methods to collect and record reliable data and information? **PC**
- Analyze, reflect on, and draw meaningful conclusions as related to the inquiry? **PA**
- Evaluate the process and results of the inquiry, troubleshooting problems if they arose? **E**
- Consider the role of scientists in innovation? **AI**
- Present the results of the inquiry using language, conventions, and representations appropriate for a specific purpose and audience? **C**